a9lim

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Independent interpretability and alignment researcher and open-source developer.

Experience

Independent Researcher & Open-Source Developer Feb 2026 - Present
a9l.im
- Conduct independent interpretability and alignment research from hypothesis through experiments and writeup; publish open-source LLM tooling.

Education

University of California, San Diego Sep 2023 - Mar 2026
B.S. Mathematics - GPA 3.75 - completed early

Selected Research

Saklas Apr 2026 - Present
Mechanistic interpretability workbench - Python, PyTorch
- Built a local workbench for probing and steering LLMs along curved manifolds.
- Implemented Jacobian-lens readouts and arbitrary-role chat support.
- Published a typed Python API, dashboard, and OpenAI/Ollama-compatible server compatible with most open source LLMs. Available on PyPI with CUDA and MPS support.

Introspection via Kaomoji Apr 2026 - Jun 2026
Emotional concept representation study - Python
- The emotional category of a prompt was decodable with 99.2% accuracy from the hidden state and the opening kaomoji alone retained 80.6% accuracy on Gemma against 11.1% by chance.
- Procrustes-aligned emotional category centroids to recover shared geometry; built kaomoji collection tooling and published the collected corpus.

Elapsed-Time Encoding in LLMs Jun 2026 - Present
Ongoing time encoding study - Python, PyTorch
- Showed elapsed conversational time is linearly encoded at an elicited token ($r = 0.88$, about 0.3 seconds per token) using linear probes.

Neuralese as a Conlang Jul 2026 - Present
Ongoing abstract-CoT analysis study - Python, PyTorch
- Implemented Abstract-CoT to test if abstract tokens could be interpreted as a compositional language.
- A preliminary 135M run did not successfully learn to use abstract tokens to reason.

Block Sparse Crosscoders Jul 2026 - Present
Ongoing cross-layer representation geometry study - Python, PyTorch
- Synthesized recent work on block sparse autoencoders and sparse crosscoders to extract cross-layer multi-dimensional features.

Selected Engineering

Geon Feb 2026 - Present
Relativistic N-body simulator - JavaScript, WebGPU
- Built a WebGPU backend to render and simulate N-body post-Newtonian physics.
- Implemented Boris integration to approximate Einstein–Infeld–Hoffmann dynamics with scalar fields.

Ogdoad Jun 2026 - Present
Research library for recreational mathematics - Rust, Python
- Built a Rust library to explore Clifford algebras over the surreal numbers and Conway's combinatorial games.
- Implemented Python bindings and designed a purpose-built language to investigate open problems.

Other Projects Feb 2026 - Present
Browser simulations and agentic tooling - JavaScript, WebGPU
- **Coding agent infrastructure.** Gaslamp, a cross-review CLI for Claude Code and Codex, and Dagwood, a version-controlled DAG-based task tracker with an MCP interface.
- **Interactive simulations.** Built seven more browser projects across physics, finance, biology, political science, and comparative religion using vanilla JS.

Technical Skills

Research/ML: mechanistic interpretability; Python, PyTorch, NumPy/SciPy, scikit-learn, CUDA/MPS.
Engineering: Java, JavaScript, WebGPU, Rust, FastAPI, Cloudflare Workers, MCP tooling, Git.